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Two-step anaerobic digestion of rice straw with nanobubble water

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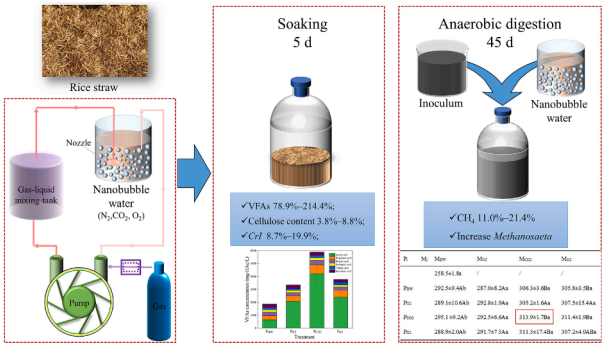
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HIGHLIGHTS

- Nanobubble water (NW) as soaking agent and AD accelerant was studied.
- Methane yield of straw with NW increased by 11.0%–21.4% compared with untreated.
- NW soaking enhanced volatile fatty acids production by 78.9%–214.4%.
- NW as AD accelerant increased relative abundance of *Methanosaeta*.

GRAPHICAL ABSTRACT



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ABSTRACT

Lignocellulose usually requires pretreatment to improve biogas production. To enhance lignocellulose biodegradability and improve anaerobic digestion (AD) efficiency, different types (N_2, CO_2, and O_2) of nanobubble water (NW) were applied in this study as soaking agent and AD accelerant to increase the biogas yield of rice straw. The results showed that the cumulative methane yields of treating with NW in two-step AD increased by 11.0%–21.4% compared with untreated straw. The maximum cumulative methane yield was 313.9±1.7 mL/gVS in straw treated with CO_2-NW as soaking agent and AD accelerant (P_{CO2}-M_{CO2}). The application of CO_2-NW and O_2-NW as AD accelerants increased bacterial diversity and relative abundance of *Methanosaeta*. This study suggested that using NW could enhance soaking pretreatment and methane production of rice straw in two-step AD; however, combined treatment with inoculum and NW or microbubble water in the pretreatment needs to compare in future.

1. Introduction

A total of 970 million tonnes of crop straw is produced annually in China, which causes serious environmental pollution and resource waste if not disposed of properly (Sun et al., 2021). Anaerobic digestion (AD) of straw is an agricultural waste treatment technology that has been widely applied, as it can simultaneously recover biogas as energy and digestate as organic fertilizer, alleviate the crisis of fossil energy

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shortages, and reduce carbon emissions (Cai et al., 2021).

To enhance the biogas production from lignocellulosic materials, many studies have been conducted focusing on pretreatments to overcome their complex and refractory lignocellulosic structure (Yu et al., 2019). At present, straw pretreatment methods are mainly physical (milling, extrusion, ultrasonic pretreatment, steam explosion pretreatment, and thermal hydrolysis), chemical (alkaline pretreatment and acid pretreatment), and biological (fungi) (Shrestha et al., 2017). Although the degradability of lignocellulose is improved after pretreatment, each type of pretreatment has its advantages and disadvantages. For example, physical pretreatment is easy to implement, but has high energy demand; chemical pretreatment has high efficiency, but is associated with pollutant or inhibitory compounds; and biological pretreatment is cost-effective, but has long pretreatment cycle (Cai et al., 2021; Guan et al., 2022; Ma et al., 2022). These limitations raise the need to explore more efficient pretreatment methods with lower energy consumption and fewer chemical additives.

Nanobubble water (NW) has defined as bubbles <1000 nm in diameter (ISO/TC281) and generated in water by mechanical stirring, hydraulic cavitation, ultrasonic cavitation, dissolved gas release, and membrane filtration etc. (Wang et al., 2021b). In recent years, NW has been widely applied in wastewater treatment (Xiao et al., 2021), membrane cleaning (Farid et al., 2022), aquaculture (Liu et al., 2013), irrigation (Zhou et al., 2022), and waste disposal (Fan et al., 2021). The interest in nanobubble is due to their unique characteristics, such as their long residence time in solution, high zeta potential compared with deionized water (DW), possible ability to generate hydroxyl radicals, and high gas-liquid mass transfer efficiency compared with microbubbles which size range is 1–100 μm (Chuenchart et al., 2021; Wang et al., 2021b). Experimental evidence has demonstrated the long-term existence of nanobubbles; however, the mechanism behind their long-term existence remains controversial.

Although the benefits of NW as AD accelerant have been confirmed, the mechanism by which NW promotes AD is unclear. Some researchers have reported that adding NW to AD reactors with waste activated sludge, pig manure, food waste, and corn straw as feedstock could increase the biogas yields by 3% – 40% compared with the addition of DW (Chuenchart et al., 2021; Wang et al., 2020b; Yang et al., 2019). These positive effects of NW addition may be attributed to improved AD performance and efficiency, which promotes substrate biodegradation (Fan et al., 2020a), prevents acid accumulation (Fan et al., 2020b), and alleviates ammonia inhibition (Fan et al., 2021). In addition, some predictions have been put forward based on research results. Negatively charged nanobubbles enhance the mobility of water molecules which is related to the mass transfer rate of nutrients from liquid to cells (Wang et al., 2019). This also means that mass transfer between microorganisms and the environment is accelerated when NW is used in an AD system. Thus, the addition of NW to an AD reactor could enhance their enzymatic activity and increase the relative abundance of major functional bacteria and archaea (Wang et al., 2020a).

There are only a few studies on the addition of NW in the AD process, while there was no comprehensive evaluation of applying NW to two-step AD process with soaking pretreatment and methane production. Therefore, the main purpose of this study was to clarify the various contributions of NW in two-step AD with soaking pretreatment and methane production. Initially, this study evaluated the effect of soaking pretreatment with different types of NW (N_2-NW, CO_2-NW, and O_2-NW) on the hydrolysis efficiency of air-dried rice straw. Following the pretreatment, the effect of NW as an AD accelerant on biogas production was also examined. In brief, the purpose of this study was to explore the effects of different types of NW on the hydrolysis performance and biogas production of air-dried rice straw with the aim of providing an efficient method for improving the biogas production of air-dried rice straw without the addition of hazardous chemical substances.

2. Materials and methods

2.1. Rice straw and inoculum

Rice straw was harvested from a farm in the Hubei Province, China (31°43′N, 112°07′E). The straw was sheared and stored in a laboratory freezer at 4 °C. Before being fed into the anaerobic reactor, the straw was cut to 1–2 cm in length using a straw chopper. The inoculum was obtained from mesophilic anaerobic digester operated stably in a municipal wastewater treatment plant in Beijing. The sludge was placed in a reactor at 37 °C for one week until no biogas was produced and then used for the batch AD experiments. The characteristics of rice straw and inoculum are shown in Table 1.

2.2. Preparation of nanobubble water

Briefly, NW was generated using N_2, CO_2, or O_2 gas (purity >99.9%, Beijing Haipu Gas Co., Ltd., China) and DW by a micro-nanobubble rapid generation device (TL-MBG10-A, Beijing Zhongnongtianlu Co., Ltd., China). NW was generated in the gas-liquid mixing tank of the generator through the hydraulic cavitation formed by the high-speed rotating shearing of the circulated water pump, and then released into the water tank through a nozzle ready for testing and use. The NW was placed for 3–5 min after preparation to make most of macro/microbubble to escape from the water, during which freshly prepared water gradually changed from milky white to transparent.

Some studies have shown that the concentration of bubbles in NW was significantly affected by the operating time of the device (Cerron-Calle et al., 2022; Wang et al., 2019). Therefore, the effect of operating time (5, 10, 15, and 20 min) on nanobubbles concentration and size distribution was explored under the settings of 1 L/min of gas intake and 20 L of water volume in the tank. For the O_2-NW and N_2-NW, the nanobubble concentration increased to $3.8\times10^8 \pm 3.7\times10^7$ and $1.43\times10^8 \pm 1.5\times10^7$ particles/mL within 15 min, respectively. The concentration became unstable (temperature of water rises as the device runs for a long time) over the following 5 min. However, for the CO_2-NW, the highest concentration of nanobubble was observed at 10 min ($1.1\times10^8 \pm 1.0\times10^7$ particles/mL). After 10 min, the concentration decreased with operation time (see supplementary material). Therefore, the operation times for generating O_2-NW and N_2-NW were 15 min, and that for generating the CO_2-NW was 10 min.

2.3. Soaking pretreatment of rice straw

To explore the effect of soaking pretreatment with different types of NW on the hydrolysis of rice straw, experiments were conducted with four types of soaking water: DW, N_2-NW, CO_2-NW, and O_2-NW, and the treatment groups were labeled as P_{DW}, P_{N2}, P_{CO2}, and P_{O2}, respectively. The solid-liquid ratio of rice straw to water was 1:10 based on the total solids (TS) in a 250-mL glass bottle. To avoid organic matter loss under aerobic conditions during the soaking process, the headspace of the bottle was flushed with N_2 for 1 min and immediately sealed with a nitrile stopper after the addition of water and rice straw. All glass bottles were placed in a constant temperature incubator (RZH-380A, Nanbei

Table 1
Characteristics of rice straw and inoculum.

Material	Rice straw	Inoculum
TS (% of wet weight)	94.3±0.3	5.4±0.1
VS (% of TS)	81.5±0.2	56.0±0.1
Cellulose (% of TS)	26.2±0.6	/
Hemicellulose (% of TS)	26.3±0.5	/
Lignin (% of TS)	4.9±0.6	/
pH	/	8.0

(1)TS, total solids; VS, volatile solids; Lignin, acid detergent lignin.
(2)The values were the mean ± standard deviation of three replicates.

Co., Ltd., China) at 37±1 °C for 5 d. All bottles were shaken by hand for 30 s every day. Fifteen bottles were included in each treatment: three bottles were used for material characteristic analysis after 5 d of soaking, and twelve bottles were used for subsequent batch AD experiments.

2.4. Set-up of batch AD experiment for rice straw

The inoculum was loaded directly into the soaking bottles with a substrate-to-inoculum based on the volatile solids (VS) ratio of was 1:1.5 after 5 days of soaking. Subsequently, different types of NW were supplemented as AD accelerants to maintain same TS in the bottles, and the treatment groups were labeled as M_{N2}, M_{CO2}, and M_{O2}, respectively. DW was also added to the AD reactor, maintaining the same TS (labeled as M_{DW}). Unsoaked rice straw supplemented with DW was used as untreated group. The biogas production of the inoculum alone was also determined (labeled as blank group) (see supplementary material). After loading, the headspace of the bottle was flushed with N_2 for 1 min and then immediately sealed. All bottles were placed at 37±1 °C in a constant temperature water bath (HH-60, Changzhou Guohua Co., Ltd., China) and shaken by hand for 30 s every day. Three replicates were set for each treatment, and the actual biogas production of the rice straw was calculated by subtracting the biogas production of the blank group from that of the treatment groups.

2.5. Analytical methods

TS and VS were measured according to the standard methods of the American Public Health Association (APHA, 2005). The pH was measured using a pH meter (FE28, Mettler-Toledo, Switzerland). Cellulose, hemicellulose, and acid detergent lignin contents were determined using a fiber analyzer (2000, Ankom Technology Corp., USA) according to Vansoest (1963). Crystallinity was measured using an X-ray diffractometer (D8, Bruker, Germany), and the relevant parameters were set according to Gao et al. (2017). The crystallinity index (*CrI*) calculation formula is shown in Eq. (1).

$$CrI(\%) = [(I_{002} - I_{am})/I_{002}] \times 100\% \quad (1)$$

where I_{002} is the maximum intensity at 2θ angles of 22°–23°, and I_{am} is the intensity of the amorphous structure in the 2θ range of 16°–18°.

The size and concentration of nanobubbles were determined using a Nanoparticle Tracking Analysis (NTA) device (NanoSight NS500, Malvern, UK), which generally provides the particle size distribution of the sample in liquid based on light scattering and Brownian motion analyses (Kestens et al., 2017). A high sensitivity CMOS camera recorded videos of the Brownian motion of the nanobubble; subsequently, their Brownian motion was analyzed by the NTA software to obtain particle size distribution and concentration. The device was flushed to ensure that there were no obvious particles by the camera before detection. The camera level and focus were adjusted to obtain the maximum image after loading each sample. Capture number was set to three times in each case.

The volume of daily biogas production was measured using a glass syringe; the syringe was inserted to the bottles, and the volume was recorded once it stabilized. The volume was subsequently normalized to the standard temperature and pressure conditions. Biogas composition was measured via gas chromatography (GC8A, Shimadzu, Japan). Volatile fatty acids (VFAs) concentrations were analyzed by gas chromatography (GC-2010 Plus, Shimadzu, Japan). The modified Gompertz model was used to analyze the kinetic characteristics of different groups of methane production. The model is shown as in Eq. (2).

$$P(t) = P_0 \times \exp\left\{ -\exp\left[\frac{R_m e}{P_0}(\lambda - t) + 1\right] \right\} \quad (2)$$

where P(t) is the cumulative methane production (mL•g-VS^{-1}); P_0 is the maximum methane potential production (mL•g-VS^{-1}); R_m is the maximum methane production rate (mL•g-VS^{-1}•d^{-1}); λ is the time of the delay phase (d); and t is the time of AD (d).

2.6. Microbial community analysis

To further explore the effect of adding NW as AD accelerant on microbial community, samples from the P_{DW}-M_{DW}, P_{DW}-M_{N2}, P_{DW}-M_{CO2}, P_{DW}-M_{O2}, and P_{CO2}-M_{CO2} treatments were collected and centrifuged for microbial community analyses. DNA was extracted using an E.Z.N.A.® Soil DNA Kit (Omega Bio-tek, Norcross, GA, USA) according to the manufacturer's protocols, while 1% agarose gel electrophoresis was used to check DNA quality. The 16S rRNA gene primers used were 338F and 806R for bacteria and 524F10extF and Arch958RmodR for archaea. The sequence was detected on Illumina MiSeq platform device (Illumina, San Diego, USA) at the Majorbio bio-Pharm Technology Co., Ltd. (Shanghai, China). Specific operational procedures and microbial community analysis methods were referred to previous studies (Sun et al., 2019). The microbial community diversity and distribution were analyzed using the Majorbio Cloud Platform (https://www.majorbio.com). The raw sequencing data were submitted to the Sequence Read Archive (SRA) of National Center for Biotechnology Information (NCBI) under project number PRJNA935990.

2.7. Data analysis

In this study, graph processing and model analysis were performed using Origin 2021 (OriginLab, USA). Significant differences were determined by one-way analysis of variance (ANOVA) using SPSS 25 software. Statistical significance was assumed at $p < 0.05$.

3. Results and discussion

3.1. Effect of soaking pretreatment on rice straw with nanobubble water

3.1.1. Volatile fatty acids production after soaking pretreatment

As shown in Fig. 1, the total VFAs concentrations in the treatments of P_{DW}, P_{N2}, P_{CO2}, and P_{O2} were 1873.3±242.5, 3351.4±117.2, 5889.6 ±928.0, and 3782.6±270.8 mg-HAc/L, respectively, after 5-days of soaking pretreatment. The results indicated that soaking pretreatment in NW significantly increased the total VFAs concentration of soaking liquid ($p < 0.05$). Particularly, acetic acid concentration significantly increased in the soaking of the NW treatment ($p < 0.05$). And the highest

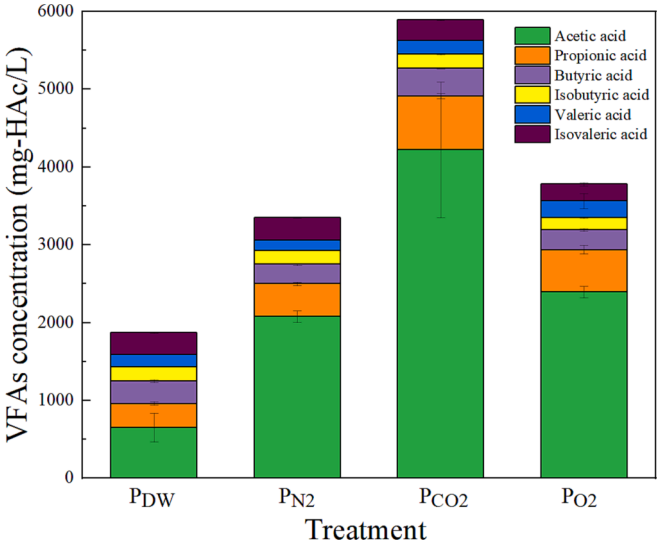


Fig. 1. VFAs concentration in the liquid after soaking pretreatment.
(1) The concentrations of VFAs are calculated as acetic acid.

acetic acid production (4221.8±870.6 mg-HAc/L) was observed in the treatment of P_{CO2}, which was 5.4-fold that in the P_{DW} treatment (655.5 ±183.0 mg-HAc/L). Likewise, acetic acid concentration reached 2079.4 ±74.5 and 2393.8±74.8 mg-HAc/L in the treatments of P_{N2} and P_{O2}, which was 2.2-fold and 2.6-fold that of the P_{DW} treatment, respectively.

To date, most studies have focused on improving hydrolysis efficiency through NW addition in the AD process. Yang et al. (2019) found that when N_2-NW was added to AD process of waste activated sludge, the VFAs concentration increased by 31.4% after the initial three days compared with the addition of DW. Wang et al. (2020a) observed that the addition of Air-NW and CO_2-NW to a high cellulose loading AD reactor increased VFAs concentrations by 30.3% and 11.4%, respectively, compared with the addition of DW. Some researchers have shown that the addition of NW changed the microscale water environment of the AD system and accelerated water mobility, which was an important factor affecting the hydrolysis process (Wang et al., 2019). With an increase in water mobility, the flow rate of substances and energy between microorganisms and the outside environment was enhanced and the release of enzymes was elevated, which was beneficial to the hydrolysis of substances (Chuenchart et al., 2021; Wang et al., 2020b). In this study, no exogenous hydrolytic bacteria were inoculated in the soaking pretreatment, and only epiphytic bacteria were present in the straw. Although no hydrolytic microorganisms were added to enhance hydrolysis in the AD process, the soaking pretreatment functioned as a pre-hydrolysis process. In this case, rice straw soaked in NW showed 78.9%–214.4% increases in VFAs production compared with that soaked in DW. Thus, coupling with NW and exogenous hydrolytic bacterial inoculum is speculated to further increase the VFAs production.

In addition, CO_2-NW exhibited unique characteristics. Compared with N_2-NW and O_2-NW, a large amount of CO_2 was dissolved into water to form carbonic acid in CO_2-NW. The low pH of CO_2-NW (i.e., 4.2) created an acidic environment which might explain why the highest VFAs production was observed in the P_{CO2} treatment (Wang et al., 2019). In addition, the increased acetic acid content in the P_{CO2} treatment would be good for the subsequent AD process, because acetic acid could be directly utilized by methanogens as the main energy source (Li et al., 2022).

3.1.2. Lignocellulose content and crystallinity index of rice straw after soaking

The contents of cellulose, hemicellulose, and acid detergent lignin in rice straw after soaking pretreatment are shown in Table 2. The cellulose content in treatments of the untreated, P_{DW}, P_{N2}, P_{CO2} and P_{O2} were 26.2% ± 0.6%, 27.2% ± 0.6%, 27.2% ± 0.5%, 28.4% ± 0.2%, and 28.5% ± 0.4%, respectively. Compared with that in the untreated group, the cellulose content increased significantly after soaking ($p < 0.05$), showing an increase of 3.8%, 3.8%, 8.4%, and 8.8%, respectively. Crystallinity was also an important factor in the hydrolysis of rice straw. The *CrI* of rice straw was 43.9%, 43.7%, 48.2%, and 44.7% in the treatments soaked in DW, N_2-NW, CO_2-NW, and O_2-NW, respectively, all of which was higher than that in the untreated group (40.2%).

These results might be due to the stronger resistance in the crystalline region of cellulose than that in the amorphous region, and the degradation of part of the amorphous cellulose and hemicellulose during soaking (Ma et al., 2022; Wei et al., 2022; Zhou et al., 2017). Therefore, part of the amorphous material was dissolved, thus destructing the complex and refractory lignocellulose structure to some extent and increasing the accessibility of the enzymes to cellulose, which was beneficial for increasing biogas production.

Table 2
Lignocellulosic content after soaking pretreatment (based on the TS).

Treatment	Cellulose (%)	Hemicellulose (%)	Lignin (%)
Untreated	26.2 ± 0.6a	26.3 ± 0.5a	4.9 ± 0.6a
P_{DW}	27.2 ± 0.6b (+3.8%)	27.1 ± 0.3b (+3.0%)	4.5 ± 0.7a (-8.2%)
P_{N2}	27.2 ± 0.5b (+3.8%)	28.0 ± 0.3c (+6.5%)	4.4 ± 0.6a (-10.2%)
P_{CO2}	28.4 ± 0.2c (+8.4%)	27.2 ± 0.3b (+3.4%)	5.5 ± 0.4a (+12.2%)
P_{O2}	28.5 ± 0.4c (+8.8%)	27.2 ± 0.4b (+3.4%)	5.1 ± 0.4a (+4.1%)

(1)Lignin represents acid detergent lignin.
(2)The values were the mean ± standard deviation of three replicates; The value in the bracket is the increase/decrease rate compared to the untreated group.
(3)a, b, and c represent the significance level of differences between samples with different pretreatments ($p < 0.05$).

3.2. Effect of methane production on rice straw with nanobubbles water

The effect of sole soaking pretreatment with different types of NW on the methane production of rice straw is illustrated in Fig. 2. Sole soaking pretreatment in either DW or NW (P_{DW}-M_{DW}, P_{N2}-M_{DW}, P_{CO2}-M_{DW}, or P_{O2}-M_{DW}) significantly increased the cumulative methane yields by 11.8%–14.2% (288.9–295.1 mL/gVS, $p < 0.05$) compared with that in the untreated group (258.5±1.8 mL/gVS). However, the improvement was comparable between the NW and the DW soaking treatment ($p > 0.05$) (Fig. 2). It was obvious that the higher peak of daily methane yield after soaking pretreatment of adding CO_2-NW and O_2-NW (P_{CO2}-M_{DW} and P_{O2}-M_{DW}) compared with DW soaking (P_{DW}-M_{DW}). As discussed in Section 3.1.1, the higher peak daily methane yield after soaking in NW could be attributed to the higher VFAs content than that after soaking in DW. Although the daily methane yield was affected by soaking pretreatment with different types of NW, the cumulative methane yields were comparable between the NW and the DW soaking treatments. Therefore, sole soaking in NW only increased the peak of the daily methane yield, but resulted in comparable the cumulative methane yield to DW.

To achieve higher biogas production, this study explored the effect of NW addition on both the soaking pretreatment and the successive AD process on the biogas production of rice straw. The cumulative methane yields matrix ($P_i \times M_j$) is presented in Table 3. For rice straw pretreated with different types of water (DW, N_2-NW, CO_2-NW, and O_2-NW) as both soaking agent and AD accelerants, the cumulative methane yield ranged from 287.0 to 313.9 mL/gVS. The P_i-M_{CO2} and P_i-M_{O2} treatments (with CO_2-NW and O_2-NW) as AD accelerants significantly increased the cumulative methane yield compared with the P_i-M_{DW} treatment ($p < 0.05$). The highest yield was obtained with the P_{CO2}-M_{CO2} treatment (313.9

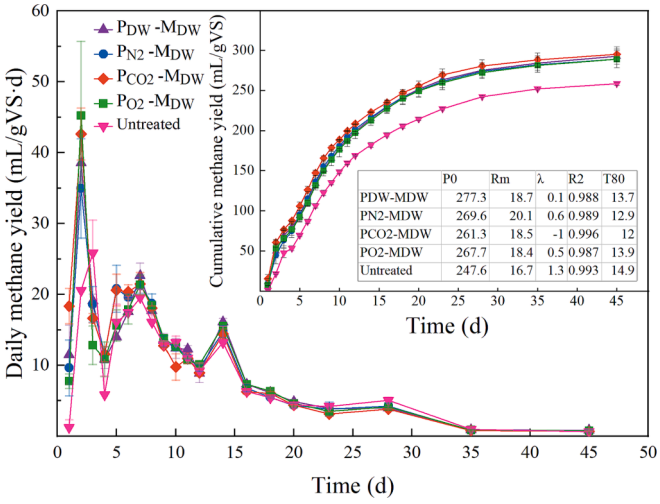


Fig. 2. Daily and cumulative methane yield affected by soaking pretreatment. (1) P_0 is the maximum methane potential production (mL•g-VS^{-1}); R_m is the maximum methane production rate (mL•g-VS^{-1}•d^{-1}); λ is the time of the delay phase (d); R^2 is correlation coefficient; T80 is digestion time of reaching 80% of P_0.

Table 3
Cumulative methane yields of rice straw treated with DW or NW as both soaking agent and AD accelerant (mL/gVS).

P_i M_j	M_{DW}	M_{N2}	M_{CO2}	M_{O2}	Significance		
					P	M	P × M
Untreated	258.5 ± 1.8a	/	/	/	/	/	/
P_{DW}	292.5 ± 9.4Ab	287.0 ± 8.2Aa	306.3 ± 3.6Ba	305.8 ± 3.5Ba	ns	***	ns
P_{N2}	289.1 ± 10.6Ab	292.8 ± 1.9Aa	305.2 ± 1.6Aa	307.5 ± 15.4Aa			
P_{CO2}	295.1 ± 9.2Ab	292.5 ± 6.6Aa	313.9 ± 1.7Ba	311.4 ± 1.9Ba			
P_{O2}	288.9 ± 2.0Ab	291.7 ± 7 3Aa	311.3 ± 17.4Ba	307.2 ± 4.0ABa			

(1)Untreated, rice straw without not soaking pretreatment; P_{DW}, soaking pretreatment with deionized water; P_{N2}, soaking pretreatment with N_2-nanobubble water (NW); P_{CO2}, soaking pretreatment with CO_2-NW; P_{O2}, soaking pretreatment with O_2-NW.
(2)M_{DW}, adding deionized water as an anaerobic digestion (AD) accelerant; M_{N2}, adding N_2-nanobubble water (NW) as an AD accelerant; P_{CO2}, CO_2-NW as an AD accelerant; P_{O2}, O_2-NW as an AD accelerant.
(3)P, soaking pretreatment; M, NW as AD accelerant; P × M, the interaction between soaking pretreatment and NW as AD accelerant; *** $p < 0.001$; ns, not significant.
(4)Lowercase letters (a and b) represent the significance level of difference between samples pretreated with different soaking pretreatment but added same NW as an AD accelerant ($p < 0.05$); Capital letters (A and B) represent the significance level of difference between samples added different NW as an AD accelerant but pretreated with same soaking pretreatment ($p < 0.05$).
(5)The values were the mean ± standard deviation of three replicates.

±1.7 mL/gVS), followed by the P_{CO2}-M_{O2} treatment (311.4±1.9 mL/gVS). The daily and cumulative methane yields of rice straw treated with different types of water (DW, N_2-NW, CO_2-NW, and O_2-NW) as both soaking pretreatment and AD accelerant are illustrated in Fig. 3. The highest daily methane yield was observed in the application of CO_2-NW as an AD accelerant ($P_i \times M_{CO2}$) within 3 days, regardless of the type of NW used as a soaking pretreatment. This meant that the addition of CO_2-NW as an AD accelerant was beneficial for the hydrolysis and acidification of organic matter.

In this study, the addition of CO_2-NW during the soaking process increased the cumulative methane yield by 14.2% compared with the untreated group without soaking. When considering only the difference between soaking pretreatment in NW and DW during soaking (P_{CO2}-M_{DW} treatment compared with P_{DW}-M_{DW} treatment), the cumulative yield increment was extremely limited (+0.9%). Interestingly, the addition of CO_2-NW during the soaking and AD processes could increase the cumulative yield by 21.4% and 7.3%, respectively, compared with the untreated group and P_{DW}-M_{DW} treatment. The increase in methane production with the addition of CO_2-NW in this study was similar to that reported in previous studies. Wang et al. (2019) found that the cumulative methane yield of waste activated sludge increased by 16% after the addition of CO_2-NW as an AD accelerant compared with DW treatment. Fan et al. (2020a) revealed that the addition of CO_2-NW as AD accelerant resulted in an increase of 20.8% in the cumulative methane yield of swine manure compared with the DW treatment. The studies also reported that a higher cumulative methane yield was achieved by adding H_2-NW, air-NW, and CO_2-NW to AD process; however, the addition of N_2-NW resulted in only a slight increase in yield. The increase in methane production might be attributed to the addition of NW as AD accelerant, which could enhance the enzyme and microbial activity of hydrolysis and methanogenesis. This study also proved that addition of NW at the soaking pretreatment and AD processes could enhance methane production and improve the efficiency of the AD system.

The highest methane production was obtained with the addition of CO_2-NW as an AD accelerant, which might be due to the provision of exogenous CO_2 as a substrate to increase methane production. In a previous study, CO_2 was injected into a sewage sludge pilot-scale AD reactor, resulting in an increase of 6% in methane yield compared with the untreated group (Muntau et al., 2021). The higher methane yield might be attributed to the stimulation hydrogenotrophic (Wang et al., 2020a) and acetoclastic (Bajón Fernández et al., 2015) methanogenesis pathways under CO_2-enriched conditions. Due to the high mass transfer efficiency of gas-liquid NW, the CO_2-NW applied in this study might lead to a greater improvement of biogas production compared with direct injection of CO_2; therefore, further research was necessary.

The modified Gompertz model was used to fit the kinetic characteristics of methane yield with different NW. The kinetic characteristics and the digestion time T80 (reaching 80% of P_0) are shown in Fig. 3 (embedded table). According to the results, all groups were fitted well (R^2 >0.984). For P_{CO2}-M_{CO2} treatment, P_0 was 303.5±4.5 mL/gVS, thereby being 22.6% and 9.4% higher than untreated (247.6±3.8 mL/gVS) and P_{DW}-M_{DW} (277.3±6.2 mL/gVS), respectively. In addition, using NW as AD accelerant decreased λ compared with DW, implying faster methane production in rice straw AD.

It is worth noting that the air-dried rice straw used in this study had a high lignocellulose content and lignification degree compared with those of freshly harvested straw, which seriously hindered the AD process (Cai et al., 2021). The improvement in methane production was not obviously evident that using sole NW soaking as a gentle pretreatment compared with DW soaking. In addition, according to previous research, the enhancement mechanism of AD process by adding NW was to enhance the activity of microorganism (Fan et al., 2020b; Wang et al., 2020b). As discussed in section 3.1.1, this study might only stimulate epiphytic microorganism in air-dry crop straw during soaking process without exogenous inoculated hydrolytic bacteria. Therefore, this study speculate that the combined application of inoculum (exogenous hydrolytic bacteria) and NW will be beneficial for the methane formation and hydrolysis.

3.3. Microbial community analysis

As discussed in Section 3.2, using NW as an AD accelerant had a greater effect on the cumulative methane yield than using as a soaking agent. Therefore, this study selected the treatment of P_{DW}-M_{DW}, P_{DW}-M_{N2}, P_{DW}-M_{CO2}, P_{DW}-M_{O2}, and P_{CO2}-M_{CO2} to analyze microbial community influenced by the NW. The Shannon index of bacteria, indicating microbial community diversity, was relatively high in the treatment using CO_2-NW or O_2-NW as an AD accelerant. This result indicated that CO_2-NW and O_2-NW could increase bacterial diversity and accelerate the rate-limiting hydrolysis step to achieve efficient biogas production (see supplementary material). This observation was confirmed by the change in the Simpson index of bacteria, which was negatively correlated with diversity. Interestingly, the Shannon index of archaea exhibited a completely opposite trend to that of bacteria. Decreased archaeal diversity was observed in the treatments using CO_2-NW or O_2-NW as an AD accelerant (see supplementary material). This could be due to the added CO_2-NW and O_2-NW in the AD system promoting the succession of microbial population (Zhang et al., 2022). It is worth noting that the decrease in archaeal diversity did not decrease the methane yield. The reasons for the decrease in archaeal diversity observed in this study need to be further studied.

The relative abundance of bacterial community abundant at the phylum level is shown in Fig. 4a. The dominant phyla in all treatments were *Firmicutes*, *Synergistota*, *Bacteroidota*, *Caldatribacteriota*, and *Chloroflexi*. *Firmicutes* was the most abundant phyla among all treatments, while it had been previously reported that *Firmicutes* played an important role in the degradation of lignocellulose to produce VFAs (Ma et al., 2022). Interestingly, in the P_{CO2}-M_{CO2} treatment, the relative abundance

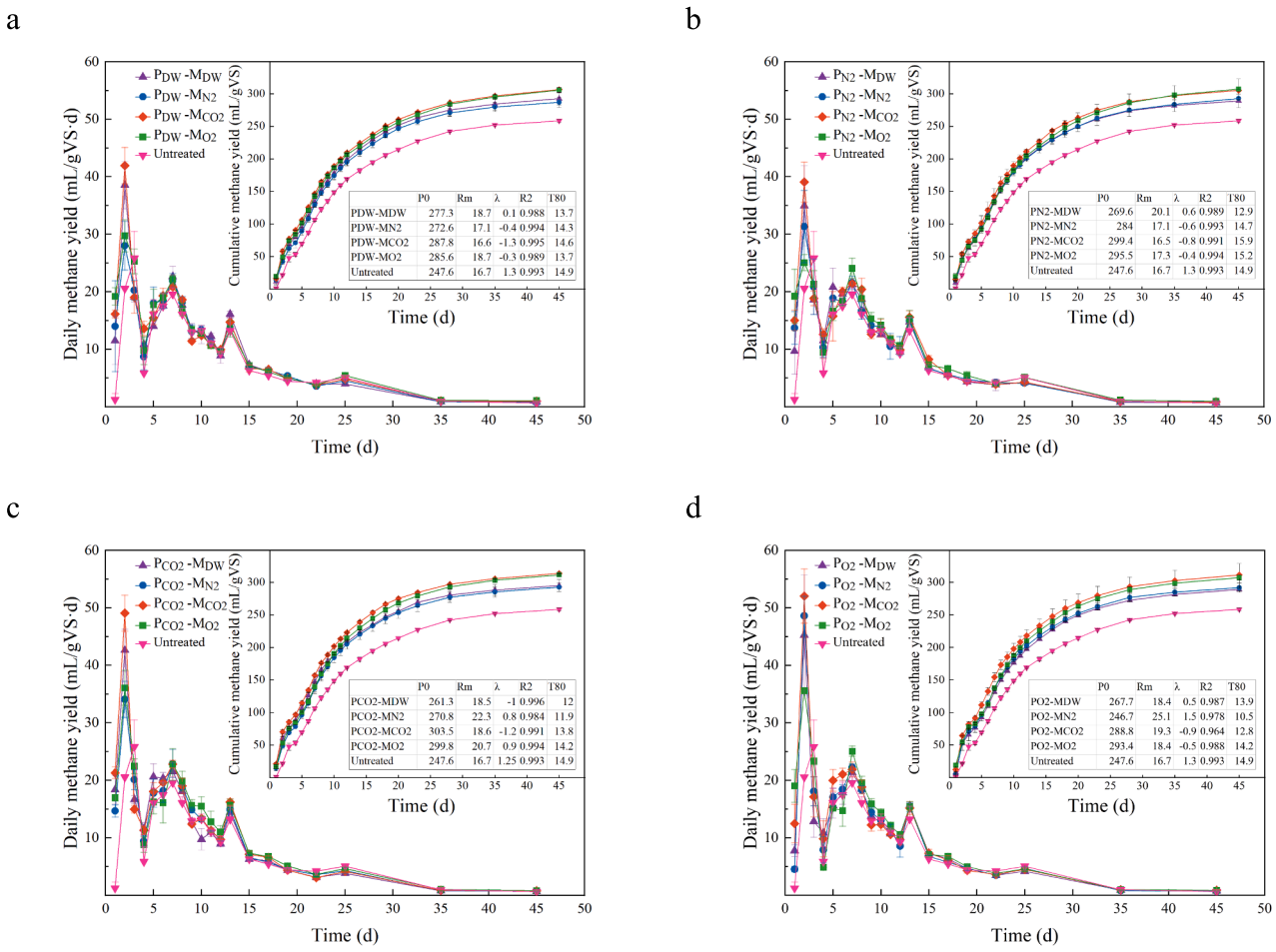


Fig. 3. Daily and cumulative methane yield affected by NW in both the soaking pretreatment and AD accelerant. (a) $P_{DW} \times M_i$; (b) $P_{N2} \times M_i$; (c) $P_{CO2} \times M_i$; (d) $P_{O2} \times M_i$.

of *Bacteroidota* and *Chloroflexi* increased compared with that in the other treatments. *Bacteroidota* had the ability to convert polysaccharides into organic acids (Wang et al., 2021a). And *Chloroflexi* was associated with the degradation of butyrate and refractory organic matter (Shao et al., 2022). It was possible that the addition of CO_2-NW during soaking (P_{CO2}-M_{CO2}) promoted the enrichment of *Bacteroidota* and *Chloroflexi*, thereby enhancing the degradation of rice straw and production of organic acid. However, compared with DW (P_{DW}-M_{DW}), adding NW as an AD accelerant (P_{DW}-M_{N2}, P_{DW}-M_{CO2}, and P_{DW}-M_{O2}) did not significantly affect the bacterial community composition.

The relative abundance of archaeal community at the genus level is shown in Fig. 4b by adding NW as an AD accelerant. The major genera were *Methanosaeta*, *Methanolinea*, *norank_c_Bathyarchaeia*, *Methanomethylovorans*, and *Methanobacterium*. The relative abundances of *Methanosaeta* in the treatment of P_{DW}-M_{DW}, P_{DW}-M_{N2}, P_{DW}-M_{CO2}, P_{DW}-M_{O2}, and P_{CO2}-M_{CO2} were 44.2%, 56.5%, 56.6%, 65.2%, and 67.3%, respectively. *Methanosaeta* was an obligate acetoclastic methanogen and was sensitive to high acetic acid concentrations in AD reactors (Ma et al., 2022; Wang et al., 2020a). This means that using NW as an AD accelerant could promote the activity of acetoclastic methanogens. The high relative abundance of *Methanolinea*, a typical hydrogenotrophic methanogen, increased after the addition of CO_2-NW, indicating the enhancement of the hydrogenotrophic methanogenesis pathway by CO_2-NW (Wang et al., 2020a). *Norank_c_Bathyarchaeia* could utilize carbohydrates, VFAs, methane, and other organic matter to produce acetyl-CoA which was subsequently consumed by *Bathyarchaeota* (Xie et al., 2022). The higher relative abundance of *Norank_c_Bathyarchaeia* might be the reason for the lower methane yield in the P_{DW}-M_{DW} treatment.

Therefore, the application of CO_2-NW and O_2-NW as AD accelerants promoted the growth of acetoclastic methanogens in this study which might be the reason for the increased biogas production observed. These results and discussion were based on the microorganisms of the sample inoculum this time; however, differences may occur under different inoculation conditions. This will be further discussed in future research on the optimization of related operating condition.

4. Conclusion

The double effect of NW as soaking agent and AD accelerant was investigated for rice straw. The biomethane yields increased by 11.0%–21.4% after the application of NW as soaking agent and AD accelerant compared with that in untreated straw. The VFAs concentration increased to 78.9%–214.4% by adding NW soaking. CO_2-NW showed the better performance compared other groups. This study suggested that using NW could enhance two-step AD process. Coupling of inoculum and NW in the hydrolysis process will be further studied to reveal the difference with traditional hydrolysis method.

CRediT authorship contribution statement

Enzhen Wang: Conceptualization, Methodology, Writing – original draft, Investigation. **Hui Sun:** Conceptualization, Writing – review & editing. **Penghui Chen:** Data curation. **Yonghui Zheng:** Investigation,

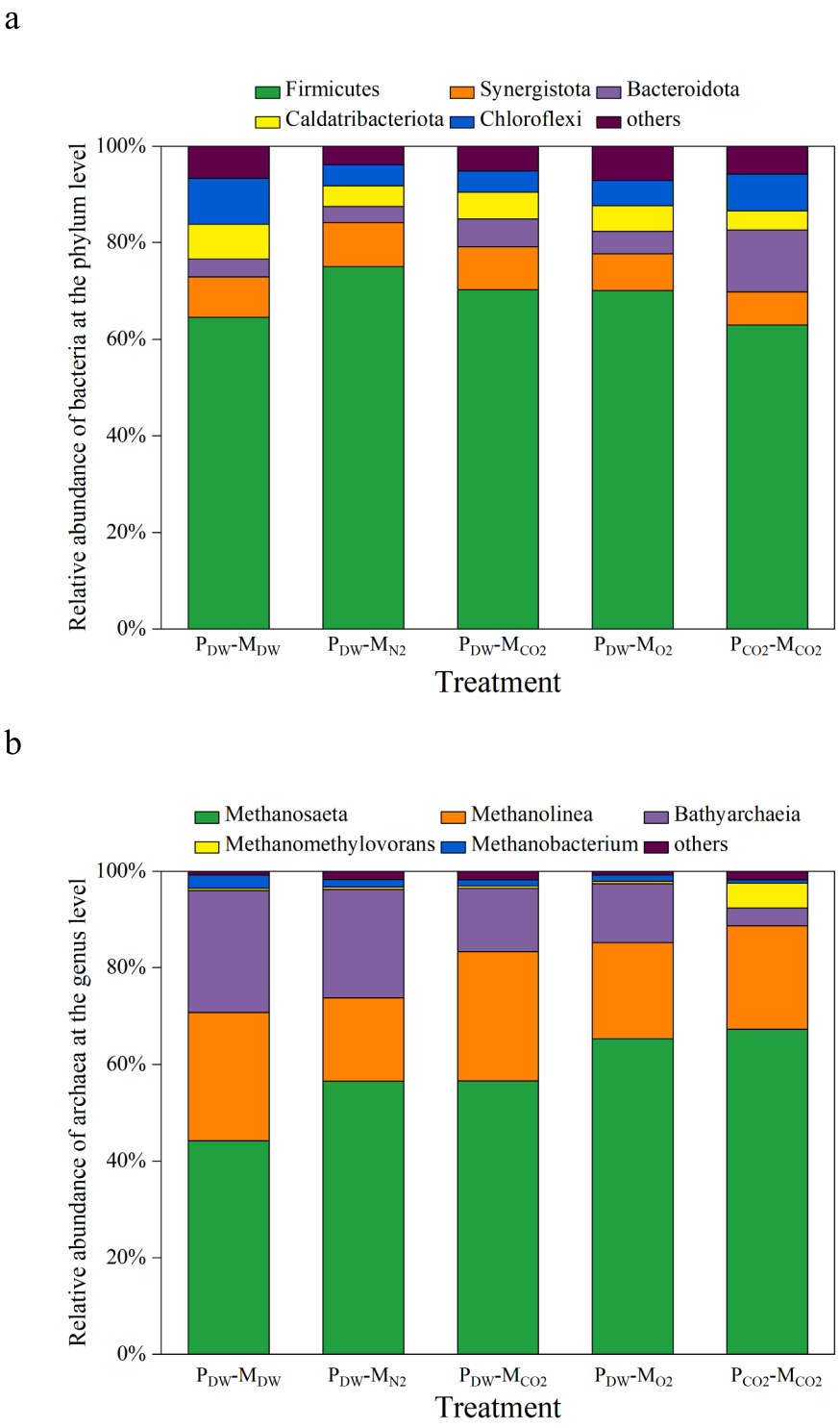


Fig. 4. Changes in the relative abundance of microbial community. (a) bacteria at the phylum level; (b) archaea at the genus level.

Visualization. **Jianbin Guo:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing, Funding acquisition. **Renjie Dong:** Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.128928.

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